

Statistical Foundation

Chapter 2

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Outline

- 1 Variables
 - Numeric Variables
 - Categorical Variables
 - Zipf's Law
- 2 Correlation Analysis
 - Analysis of Variance (ANOVA)
- 3 Statistical Significance
 - T-Value

1. Variables

Variable

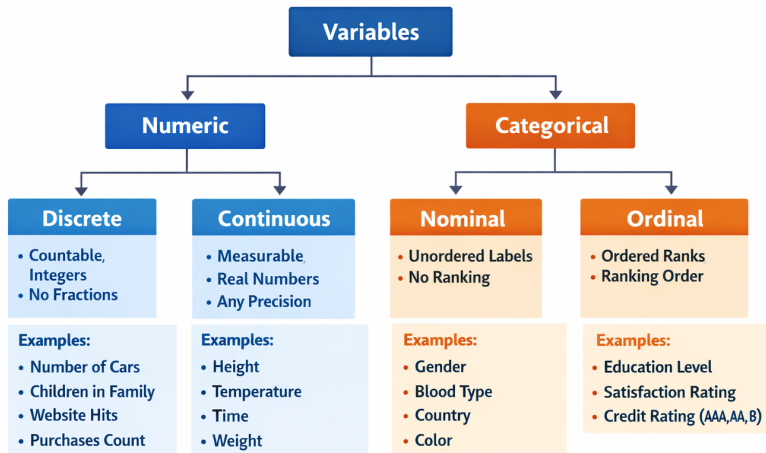
- any characteristic, number, or quantity that can be measured or counted
- variable: value can vary from one data point to another
 - your age is different from your neighbor's age

Variable	Possible Values	Type
Customer ID	1001, 1002, 1003, ...	Identifier
Age	25, 32, 41, 58, ...	Numeric
Income	45000, 75000, 120000	Numeric
Gender	Male, Female, Other	Categorical
Education	HS, BS, MS, PhD	Categorical (Ordinal)
Satisfaction	1, 2, 3, 4, 5	Categorical (Ordinal)

Two main categories:

- Numeric/Quantitative
- Categorical/Qualitative

Variable Classification Hierarchy



Numeric Variables

Numeric Variables (Quantitative)

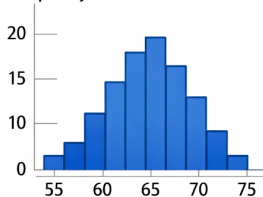
- represents measurable quantities
 - is expressed in numbers
- perform mathematical operations (like addition or averaging) on them
- Types:
 - a Continuous Variables
 - b Discrete Variables

Continuous Variables:

- can take any value within a range
- usually measurements that can be broken down into smaller decimals
 - No gaps (smooth flow) between values
- Measurable: requires measurement instrument, not counted
 - Height: 175.3 cm
- Precision limited by measurement precision
 - Weight: 70.123 kg
- Visualization: Histograms or Line Graphs or Scatter Plot
- Examples: Height (175.5 cm), Temperature (22.4°C), Time (12.5 seconds)

Continuous Variables: Visualization

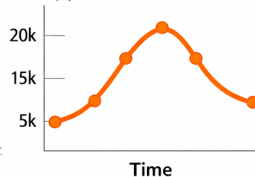
Frequency



Histogram Example

Height Distribution

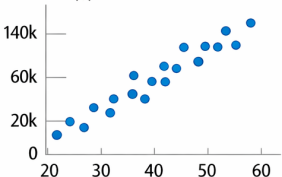
Sales (\$)



Line Graph Example

Sales Trend Over Time

Income (\$)

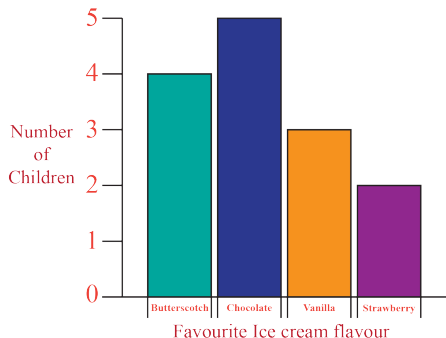


Scatter Plot Example

Age vs. Income

Discrete Variables

- represent counts
- can only take on specific, distinct values (usually whole numbers) and have “gaps” between them
- Visualization: Bar chart or probability mass function (gaps between bars)



- Examples: Number of Children: 2 or 3 but never 2.5, Dice Rolls: 4 or 5

Categorical Variables

Categorical Variables

- Values representing groups or labels or descriptions with no intrinsic ordering or ranking
 - describe a **quality** rather than a **quantity**
- Cannot perform arithmetic operations
- Mode is the only measure of central tendency
- Types:
 - a Nominal Variables
 - b Ordinal Variables

Nominal Variables

- Categories that have no logical order or ranking without quantitative meaning
 - just labels
 - 0 for Male, 1 for Female
- Categories are mutually exclusive
 - no data point can be in two categories at once
 - blood type (A, B, AB, O)
- Operations: $=$, $\neq$, Mode, Frequency counts
- Examples: Eye Color (Blue, Brown, Green), Gender (Male, Female), Country of Birth

Ordinal Variables

- categories with a meaningful order or ranking
 - intervals between categories are not necessarily equal
- Natural ordering exists ($A < B < C$)
- Differences between levels are not uniform
- Operations: $=$, $\neq$, Median, Mode, Rank comparisons, Percentiles
- Examples: Survey Satisfaction (Poor, Satisfactory, Good, Excellent), Education Level (High School, Bachelors, Masters, PhD)

Variable Type Decision Tree

Start Here:

- Does the variable represent numbers with arithmetic meaning?
 - **Yes** → **Numeric**
 - Can values be fractions?
 - Yes → **Continuous** (Examples: Height, Weight, Time)
 - No → **Discrete** (Examples: Number of Cars, Number of Kids, Counts)
 - **No** → **Categorical**
 - Is there a natural order?
 - Yes → **Ordinal** (Examples: Education Level, Rating, Rank)
 - No → **Nominal** (Examples: Gender, Country, Color)

Quick Reference:

- Can I calculate an average? → Probably **Numeric**
- Does order matter? → If YES → **Ordinal**, if NO → **Nominal**
- Are there decimal values? → If YES → **Continuous**

Encoding Variables for Analysis

Why Encode?

- Most machine learning algorithms require numeric input

Encoding Methods by Type

Variable Type	Encoding Method	Example
Nominal	One-Hot Encoding	Gender: M $\rightarrow$ [1, 0], F $\rightarrow$ [0, 1]
Ordinal	Ordinal Encoding (preserve order)	Education: HS $\rightarrow$ 1, BS $\rightarrow$ 2, MS $\rightarrow$ 3, PhD $\rightarrow$ 4
Numeric (Discrete)	No encoding needed	Age: 25, 32, 41
Numeric (Continuous)	Normalization may be needed	Height: 175.3 cm

Example: One-Hot Encoding

Original Color: Red | Green | Blue |

	Red	Green	Blue
--	-----	-------	------

	1	0	0
Encoded	0	1	0
	0	0	1
	1	0	0

Original Gender: M | F |

	M	F
--	---	---

Encoded	1	0
	0	1

Binomial Distribution

- In n independent trials, the exact probability of achieving exactly k successes is given by the Probability Mass Function (PMF):

$$P(k; n, p) = \binom{n}{k} p^k (1 - p)^{n-k}$$

- $\binom{n}{k}$ (The Binomial Coefficient): Calculated as $\frac{n!}{k!(n-k)!}$
 - represents the total number of different combinations or ways you can arrange k successes across n trials
 - p^k : probability of getting k successes
 - $(1 - p)^{n-k}$: The probability of getting the remaining failures $(n - k)$.

Statistical Metrics:

- Expected Value / Mean (μ): $\mu = np$ (The average number of successes expect to see)
- Variance (σ^2): $\sigma^2 = np(1 - p)$
- Standard Deviation (σ): $\sigma = \sqrt{np(1 - p)}$

Application

- E-Commerce Churn Prediction: A customer either stays with a subscription service (0) or cancels/churns (1)
- Credit Risk Scoring: A bank evaluates a loan application to predict if the borrower will fully repay the loan (0) or default (1)
- Medical Diagnostic Data: A patient testing kit yields a result that is either negative (0) or positive (1)
- Email spam detection: A received email can be either ham/ normal (0) or spam (1)

Zipf's Law

Zipf's Law

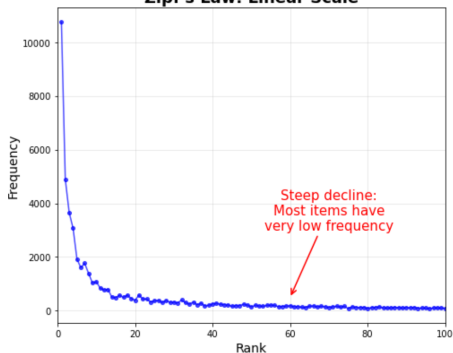
- empirical law that describes the frequency distribution of words in a natural language
 - rank-frequency relation in many types of datasets
- frequency of any word is inversely proportional to its rank in the frequency table
 - most frequent word occurs N times, the second most frequent word will occur roughly $\frac{N}{2}$ times, the third most frequent word will occur $\frac{N}{3}$ times, and the r -th most frequent word will occur $\frac{N}{r}$ times
- Zipf's Law is a discrete power law distribution
 - Let $P(r)$ be the frequency (or probability) of an item with rank r :

$$P(r) \propto \frac{1}{r^s}$$

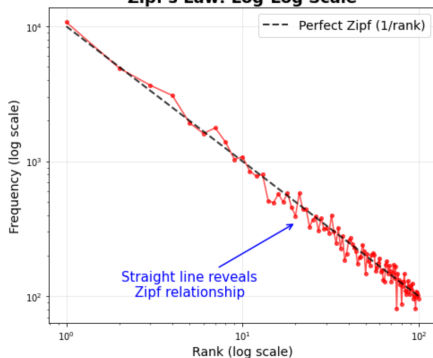
- r : rank of the item (Rank 1 is the most frequent, Rank 2 is second, etc.)
- s : scaling exponent that characterizes the distribution
 - pure natural language text, $s \approx 1$

Example

Zipf's Law: Linear Scale



Zipf's Law: Log-Log Scale



Word Frequencies

In a large English corpus:

- *the*: most common word (rank 1)
 - the appears 10,000 times
- *of*: second (rank 2)
 - of will appear around 5,000 times
- *and*: third (rank 3)
 - and will appear around 3,333 times

Application

- Natural Language Processing (NLP): Word frequencies count
- Website Traffic: Popular websites receive disproportionately high traffic
- Social Networks: A few users have extremely large followers

2. Correlation Analysis

Analysis of Variance (ANOVA)

Analysis of Variance (ANOVA)

- compare the mean of a continuous variable across two groups
- compares the variance between groups to the variance within groups to determine if the group means are statistically different

Types of ANOVA

- One-way ANOVA: Compares means across one independent variable with three or more levels
- Two-way ANOVA: Examines the effect of two independent variables

One-way ANOVA

- Used to compare means of the numeric variable across multiple categories
 - three or more independent (unrelated) categories ($k \geq 3$)
 - **Dependent Variable** single, continuous numerical feature

Hypotheses

- Null Hypothesis (H_0): $\mu_1 = \mu_2 = \mu_3 = \dots = \mu_k$
 - means of all groups are identical
 - independent variable: zero effect
- Alternative Hypothesis (H_1): At least one population mean (μ_j) is different from the others
 - does not state that all groups are different from one another
 - only that at least one pair contains a significant dislocation

One-way ANOVA

- Used to compare means of the numeric variable across multiple categories
 - three or more independent (unrelated) categories ($k \geq 3$)
 - **Dependent Variable** single, continuous numerical feature

F-Statistic Formula: compare two variances

$$F = \frac{\text{Variance BETWEEN Groups}}{\text{Variance WITHIN Groups}} = \frac{MS_{\text{between}}}{MS_{\text{within}}}$$

MS_{between}

- **Mean square between groups:** measures how much the individual group means deviate from the overall grand mean of the entire dataset

MS_{within}

- **Mean square within groups:** measures how much individual data points deviate from their own group's mean

One-way ANOVA

- Used to compare means of the numeric variable across multiple categories
 - three or more independent (unrelated) categories ($k \geq 3$)
 - **Dependent Variable** single, continuous numerical feature

Fundamental Assumptions

- Normality: continuous dependent variable $\rightarrow$ normally distributed within each of the groups
- Homogeneity of Variance (Homoscedasticity): variance of the data points within each group must be approximately equal across all groups
- Independence: observations must be collected independently of one another

Source of Variation	SS	df	MS
Between Groups	SS_{between}	$k - 1$	$MS_{\text{between}} = \frac{SS_{\text{between}}}{k - 1}$
Within Groups	SS_{within}	$N - k$	$MS_{\text{within}} = \frac{SS_{\text{within}}}{N - k}$
Total	SS_{total}	$N - 1$	

$$F = \frac{MS_{\text{between}}}{MS_{\text{within}}}$$

SS: *Sum of Squares*; **df:** *Degrees of Freedom*; **MS:** *Mean Square*

k is the number of groups, and N is the total number of samples across all groups combined

$$SS_{\text{total}} = SS_{\text{between}} + SS_{\text{within}}$$

$$SS_{\text{between}} = \sum_{j=1}^k n_j (\bar{x}_j - \bar{x}_{\text{grand}})^2$$

- k : total number of groups
- n_j : sample size (number of observations) in group j
- $\bar{x}_j$: sample mean of group j
- $\bar{x}_{\text{grand}}$: Grand mean: average of all data points across all groups combined

$$SS_{\text{within}} = \sum_{j=1}^k \sum_{i=1}^{n_j} (x_{ij} - \bar{x}_j)^2$$

$$SS_{\text{total}} = \sum_{j=1}^k \sum_{i=1}^{n_j} (x_{ij} - \bar{x}_{\text{grand}})^2$$

- x_{ij} : individual data point i located inside group j

Eta-Squared (Effect Size)

$$\eta^2 = \frac{SS_{\text{between}}}{SS_{\text{total}}} \quad \text{Range: } 0 \leq \eta^2 \leq 1$$

- Measures proportion of variance explained by group differences.
- Higher values $\rightarrow$ stronger effect.

Post-Hoc Testing (What happens after an ANOVA?)

$p\text{-value} < 0.05$, the test tells:

- At least one of these groups is statistically different
- Does not tell which specific group
- conduct post hoc tests (Tukey's HSD) to identify which specific groups differ

Example

ID	Education Level (X)	Annual Income (Y)
1	High School	38,000
2	High School	42,000
3	Bachelors	55,000
4	Bachelors	62,000
5	Bachelors	68,000
6	Masters	82,000
7	Masters	78,000
8	Masters	95,000
9	PhD	115,000
10	PhD	130,000

Example

Step 1: State hypotheses

- $H_0 : \mu_1 = \mu_2 = \mu_3 = \mu_4$ (all group means equal)
- H_1 : At least one group mean differs

Step 2: Organize data by group

$n_1 = 2, n_2 = 3, n_3 = 3, n_4 = 2$, Total $N = 10$

Group 1	High School	38,000 ; 42,000
Group 2	Bachelors	55,000 ; 62,000 ; 68,000
Group 3	Masters	82,000 ; 78,000 ; 95,000
Group 4	PhD	115,000 ; 130,000

Example

Step 3: Compute group means

$$\bar{X}_1 = \frac{38000 + 42000}{2} = \frac{80000}{2} = 40000$$

$$\bar{X}_2 = \frac{55000 + 62000 + 68000}{3} = \frac{185000}{3} \approx 61666.67$$

$$\bar{X}_3 = \frac{82000 + 78000 + 95000}{3} = \frac{255000}{3} = 85000$$

$$\bar{X}_4 = \frac{115000 + 130000}{2} = \frac{245000}{2} = 122500$$

Step 4: Grand Mean

$$\sum X = 80000 + 185000 + 255000 + 245000 = 765000$$

$$\bar{X}_{\text{grand}} = \frac{765000}{10} = 76500$$

Example

Step 5: Total sum of squares (SS_{total})

$$SS_{\text{total}} = \sum_{j=1}^k \sum_{i=1}^{n_j} (x_{ij} - \bar{X}_{\text{grand}})^2 = \sum_{j=1}^N (X_j - \bar{X}_{\text{grand}})^2$$

- **Group 1 (High School):**

$$(38000 - 76500)^2 + (42000 - 76500)^2 = 2,672,500,000$$

- **Group 2 (Bachelors):**

$$(55000 - 76500)^2 + (62000 - 76500)^2 + (68000 - 76500)^2 = 744,750,000$$

- **Group 3 (Masters):**

$$(82000 - 76500)^2 + (78000 - 76500)^2 + (95000 - 76500)^2 = 374,750,000$$

- **Group 4 (PhD):**

$$(115000 - 76500)^2 + (130000 - 76500)^2 = 4,344,500,000$$

$$\begin{aligned} SS_{\text{total}} &= 2,672,500,000 + 744,750,000 + 374,750,000 + 4,344,500,000 \\ &= 8,136,500,000 \end{aligned}$$

Example

Step 6: Between-Groups Sum of Squares (SS_{between})

$$SS_{\text{between}} = \sum_{j=1}^k n_j (\bar{x}_j - \bar{X}_{\text{grand}})^2$$

$$\text{Group 1: } 2 \times (40000 - 76500)^2 = 2,664,500,000$$

$$\text{Group 2: } 3 \times (61666.67 - 76500)^2 = 660,083,331$$

$$\text{Group 3: } 3 \times (85000 - 76500)^2 = 216,750,000$$

$$\text{Group 4: } 2 \times (122500 - 76500)^2 = 4,232,000,000$$

$$\begin{aligned} SS_{\text{between}} &= 2,664,500,000 + 660,083,331 + 216,750,000 + 4,232,000,000 \\ &= 7,773,333,331 \end{aligned}$$

Step 7: Within SS

$$\begin{aligned} SS_{\text{within}} &= SS_{\text{total}} - SS_{\text{between}} \\ &= 8,136,500,000 - 7,773,333,331 = 363,166,669 \end{aligned}$$

Example

Step 8: Degrees of Freedom

$$df_{\text{between}} = k - 1 = 4 - 1 = 3$$

$$df_{\text{within}} = N - k = 10 - 4 = 6$$

$$df_{\text{total}} = N - 1 = 9$$

Step 9: Mean Squares

$$MSB = MS_{\text{between}} = \frac{SS_{\text{between}}}{df_{\text{between}}} = \frac{7,773,333,331}{3} \approx 2,591,111,110.33$$

$$MSW = MS_{\text{within}} = \frac{SS_{\text{within}}}{df_{\text{within}}} = \frac{363,166,669}{6} \approx 60,527,778.17$$

Step 10: F-ratio

$$F = \frac{MSB}{MSW} = \frac{2,591,111,110.33}{60,527,778.17} \approx 42.81$$

Example

Step 11: Effect Size: Eta-Squared (η^2)

$$\eta^2 = \frac{SS_{\text{between}}}{SS_{\text{total}}} = \frac{7,773,333,331}{8,136,500,000} \approx 0.9554$$

95.5% of the variance in annual income is explained by education level → very large effect

Step 12: p-value

- $\alpha = 0.05$, $df_1 = 3$, $df_2 = 6$
- Critical $F \approx 4.76$ (from F-table)
- $F = 42.81 > 4.76 \rightarrow \text{Reject } H_0$
- $p < 0.0002$ (very small)

Example

Finding

strong statistical evidence: *mean annual income differs significantly across education levels*

Next Step

Post-hoc tests (e.g., Tukey HSD) would be needed to see which specific groups differ

Problem: A researcher wants to study if there's a relationship between teaching method and student exam scores

Traditional	Online	Hybrid
75	68	82
78	72	85
72	65	88
80	70	79
76	69	90

Test if there's a significant difference in mean scores across teaching methods at $\alpha = 0.05$

Feature Selection: Numerical vs. Categorical Screening

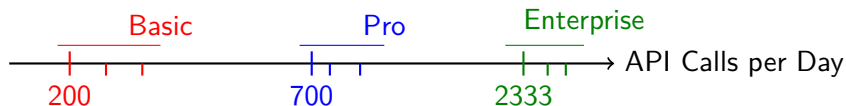
Programmatically screen **continuous features** against a **categorical target variable**

The Simulation Matrix ($N = 10$):

- **Target Class (Y):** Subscription Tier (Basic, Pro, Enterprise)
- **Numerical Feature (X):** API Calls per Day

Class	Sample API Calls	Mean
Basic	180, 200, 220	200
Pro	680, 700, 720	700
Enterprise	2200, 2333, 2400	2333.3

Feature Selection: Numerical vs. Categorical Screening



- Massive between-group separation
- Tight internal group clustering

Statistical Result

→ High F-Score / Low p-value

- variable offers excellent class-separation boundaries for classification models
- good feature = **low within-group variance** + **high between-group variance**
- ANOVA F-test quantifies this separation
- Small p-value → feature helps predict the target

3. Statistical Significance

Statistical Significance

mathematical framework indicating that an observed result, difference, or relationship in data is likely genuine and not just a fluke caused by random chance

- assuming the null hypothesis is true

Goal: whether to **reject** H_0 or **fail to reject** H_0

Decision Metrics: Test Statistics (t, χ^2, F):

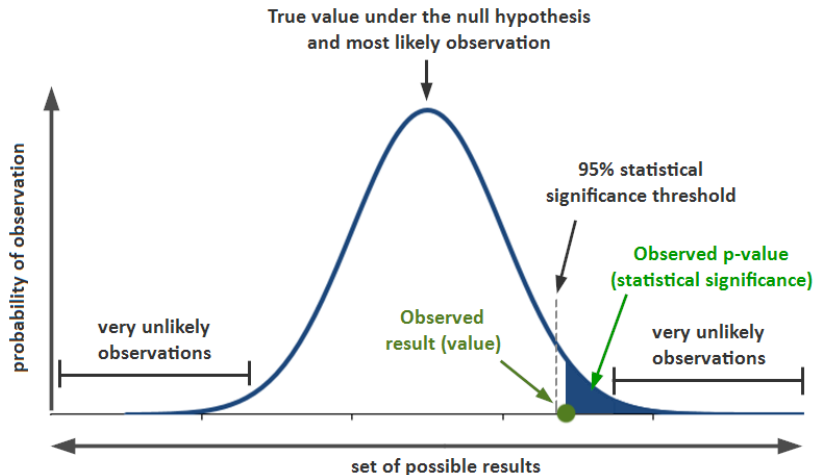
- Mathematical values tracking how far sample deviates from H_0
- p -value: The probability of seeing a result this extreme if the null hypothesis is completely true

Significance Level (α)

- **significance level** (α) is the probability of rejecting the null hypothesis when it is actually true
- Set threshold (α) before conducting the test
 - $\alpha = 0.05$ (5% risk of false positive), 0.01 (1%)
- If $p\text{-value} < \alpha$, result is statistically significant

	$p < \alpha$	$p \geq \alpha$
Decision	Reject H_0	Fail to reject H_0
Conclusion	Significant	Not significant
Interpretation	Evidence for H_1	Insufficient evidence
Action	Keep Feature	Drop/Review Feature

Significance Level (α)



"Not significant" does NOT mean "no effect" — means lack sufficient evidence

Common Significance Levels

α Level	Confidence Level	Strictness	Common Use
0.10	90%	Lenient	Exploratory research
0.05	95%	Standard	Most scientific studies
0.01	99%	Strict	Medical/clinical trials
0.001	99.9%	Very Strict	Particle physics

Interpretation

$\alpha = 0.05$ means: 5% of the time, we will make a False Positive error (Type I Error)

- Probability of making this mistake $\alpha = 0.05 = \frac{1}{20}$

One-Way ANOVA Example

- Test statistic: $F = 42.81$
- Degrees of freedom: $df_1 = 3, df_2 = 6$
- Critical value at $\alpha = 0.05$: $F_{\text{crit}} = 4.76$
- $p\text{-value} < 0.0002$

Conclusion

Since $p < 0.05$, we reject the null hypothesis.
The result is statistically significant.

Confidence Interval $(1 - \alpha)$

- actual range of values derived from the sample data

$$\text{Confidence Level} = 1 - \alpha$$

$$\text{Confidence Level} + \alpha = 1$$

P-value (probability value)

- probability of obtaining test results at least as extreme as the observed results, assuming the null hypothesis is true

$$p = P(\text{data} | H_0)$$

- Boundary: calculated p -value is compared directly against the pre-set significance level threshold (α , typically 0.05)
- Small p -value ($p < \alpha$): **Action: Reject H_0**
 - Conclusion: Statistically Significant
 - Strong evidence against the null; the observed effect is likely genuine
- Large p -value ($p \geq \alpha$): **Action: Fail to Reject H_0**
 - Conclusion: Not Statistically Significant
 - Insufficient evidence to rule out pure random chance

Misconception

Misconception 1: p -value: probability that the null hypothesis is true

- probability of the data given the hypothesis, not the probability of the hypothesis given the data

$$P(\text{data}|H_0), \text{ not } P(H_0|\text{data})$$

- Given: *If it is raining, the grass is wet.*

$$\Rightarrow P(\text{Wet Grass}|\text{Rain}) = 100\% \not\Rightarrow P(\text{Rain}|\text{Wet Grass}) = 100\%$$

Misconception 2: smaller p -value means a larger, more important real-world effect

- only measures the certainty of an effect's existence

Reality: It . A tiny effect in a massive sample size can yield a microscopic p -value. **Misconception 3:** p -value is an absolute, immutable property of the data. Reality: It is highly sensitive to external constraints, including your total sample size (N) and the structural design of your statistical test.

Statistical Significance vs Practical Significance

Statistical Significance	Practical Significance
"Is the effect real?"	"Does the effect matter?"
Based on p-value	Based on effect size (η^2 , Cohen's d)
Affected by sample size	Not affected by sample size
Tells us to reject H_0	Tells us magnitude of difference

"A statistically significant result can be practically meaningless, especially with large samples."

T-Value

T-value in Student's *t*-tests

- numerical output generated by a Student's *t*-test
 - inferential statistical test that determines whether there is a significant difference between the means of two groups
 - specifically when the population variance is unknown
- evaluate the relationship between a continuous numerical variable and a binary categorical variable
 - smaller sample sizes ($n < 30$)

One-Sample *t*-test

$$t = \frac{\text{Difference Between Group Means}}{\text{Variability of the Groups}} = \frac{\bar{x} - \mu_0}{\frac{s}{\sqrt{n}}}$$

- $\bar{x}$ = Sample mean (what we observed)
- μ_0 = Hypothesis population mean under H_0 (what we expected)
- s = Sample standard deviation (variability in our data)
- n = Sample size (number of observations)

T-value in Student's t-tests

- numerical output generated by a Student's t -test
 - inferential statistical test that determines whether there is a significant difference between the means of two groups
 - specifically when the population variance is unknown
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 - smaller sample sizes ($n < 30$)

One-Sample t-test

$$t = \frac{\text{Difference Between Group Means}}{\text{Variability of the Groups}} = \frac{\bar{x} - \mu_0}{\frac{s}{\sqrt{n}}}$$

- denominator (Standard Error) is very small, it means your data is very consistent
- Larger $|t|$ values: observed mean is far from the hypothesis mean relative to the background variability
 - strong evidence to reject the null hypothesis
- Degrees of Freedom $df = n - 1$

Example

Calculating the t-value on testing a new server algorithm where Old algorithm mean ($\mu_0 = 50$ sec) and New algorithm sample $\bar{x} = 47$ sec, $s = 6$ sec, $n = 25$. Assume significance level $\alpha = 0.05$.

$$t = \frac{\bar{x} - \mu_0}{\frac{s}{\sqrt{n}}} = \frac{47 - 50}{6/\sqrt{25}} = -2.5$$

$$df = n - 1 = 25 - 1 = 24$$

Find Critical Value: Look up t_{critical} in the t-table for $df = 24$, $\alpha = 0.05$ (Two tailed $\rightarrow \pm 2.064$)

Compare: $|t| > t_{\text{critical}} \rightarrow 2.5 > 2.064$

Conclusion: Reject H_0 . The difference is statistically significant:
"confidently say the new algorithm is statistically faster"

Example

Given sample data [3.2, 3.5, 3.7, 3.9, 4.0] and population mean $\mu_0 = 3.0$, calculate the t-statistic, assuming significance level $\alpha = 0.05$.

Calculate the sample mean ($\bar{x}$):

$$\bar{x} = \frac{\sum x_i}{n} = \frac{3.2 + 3.5 + 3.7 + 3.9 + 4.0}{5} = \frac{18.3}{5} = 3.66$$

Calculate the sample standard deviation (s):

$$\begin{aligned} s &= \sqrt{\frac{\sum (x_i - \bar{x})^2}{n - 1}} = \sqrt{\frac{0.4356 + 0.1296 + 0.0256 + 0.0016 + 0.0196}{4}} \\ &= \sqrt{\frac{0.412}{4}} = \sqrt{0.103} \approx 0.321 \end{aligned}$$

Calculate the t-statistic:

$$t = \frac{\bar{x} - \mu_0}{\frac{s}{\sqrt{n}}} = \frac{3.66 - 3.0}{\frac{0.391}{\sqrt{5}}} = \frac{0.66}{0.1749} \approx 4.597$$

Example

Degrees of freedom (df):

$$df = n - 1 = 5 - 1 = 4$$

Two-tailed p -value:

$$0.005 < p\text{-value} < 0.01$$

Interpretation: Since $p < 0.05$, reject the null hypothesis (H_0) at the $\alpha = 0.05$ significance level.

Two-Sample t-Test

statistical test used to determine whether the population means of two completely independent, unrelated groups are significantly different from one another

$$t = \frac{\text{Difference Between Group Means}}{\text{Variability of the Groups}} = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$$

Degrees of Freedom (df)

$$df = n_1 + n_2 - 2$$

Assuming Equal Variances

$$s_p^2 = \frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}$$

T-Statistic:

$$t = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{s_p^2 \left(\frac{1}{n_1} + \frac{1}{n_2} \right)}}$$

Example

You are a Data Engineer testing two different server architectures for a high-traffic web application. You want to know if there is a statistically significant difference in their mean response times. Assuming significance level $\alpha = 0.05$ and equal variances

Metric	Architecture A	Architecture B
Sample Size (n)	$n_1 = 15$	$n_2 = 15$
Sample Mean ($\bar{x}$)	$\bar{x}_1 = 120$ ms	$\bar{x}_2 = 110$ ms
Sample Std Dev (s)	$s_1 = 12$ ms	$s_2 = 10$ ms

Step 1: Calculate Pooled Variance (s_p^2)

$$s_p^2 = \frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2} = \frac{(15 - 1)(12)^2 + (15 - 1)(10)^2}{15 + 15 - 2} = 122$$

Example

Step 2: Calculate the Standard Error

$$SE = \sqrt{s_p^2 \left(\frac{1}{n_1} + \frac{1}{n_2} \right)} = \sqrt{(122)^2 \left(\frac{1}{15} + \frac{1}{15} \right)} \approx 4.033$$

Step 3: Calculate the T-Value

$$t = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{s_p^2 \left(\frac{1}{n_1} + \frac{1}{n_2} \right)}} = \frac{120 - 110}{4.033} \approx 2.479$$

Step 4: Determine Degrees of Freedom (df)

$$df = n_1 + n_2 - 2 = 15 + 15 - 2 = 28$$

Example

Step 5: Find the Critical Value

Using a t-distribution table for a two-tailed test at $\alpha = 0.05$ and $df = 28$

Critical t-value: ± 2.048 and $|t| = 2.479$

Compare:

- $2.479 > 2.048$, t-value falls in the rejection region
- Reject the Null Hypothesis (H_0)
- Architecture B is significantly faster than Architecture A
 - As a data engineer, you have statistical evidence to recommend migrating to Architecture B.

Example

Given Sample 1 [10,12,14,15,16] and Sample 2 [11,12,13,14,15], calculate the t-statistic, assuming equal variances with significance level $\alpha = 0.05$.

Hypotheses

- **Null Hypothesis (H_0):** $\mu_1 = \mu_2$ (Group means are equal)
- **Alternative Hypothesis (H_1):** $\mu_1 \neq \mu_2$ (Group means are different)

Step 1: Calculate Sample Means & Variances

Sample 1:

$$\bar{x}_1 = \frac{10 + 12 + 14 + 15 + 16}{5} = 13.4, \quad s_1^2 = \frac{\sum (x_{i1} - \bar{x}_1)^2}{5 - 1} = 5.75$$

Sample 2:

$$\bar{x}_2 = \frac{11 + 12 + 13 + 14 + 15}{5} = 13.0, \quad s_2^2 = \frac{\sum (x_{i2} - \bar{x}_2)^2}{5 - 1} = 2.5$$

Example

Step 2: Calculate Pooled Variance (s_p^2)

$$s_p^2 = \frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2} = \frac{(4 \times 5.75) + (4 \times 2.5)}{5 + 5 - 2} = \frac{33}{8} = 4.125$$

Step 3: Compute the t -Statistic Ratio

$$t = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{s_p^2 \left(\frac{1}{n_1} + \frac{1}{n_2} \right)}} = \frac{13.4 - 13.0}{\sqrt{4.125 \left(\frac{1}{5} + \frac{1}{5} \right)}} = \frac{0.4}{\sqrt{1.65}} \approx 0.312$$

Step 4: Degrees of Freedom (df): $n_1 + n_2 - 2 = 5 + 5 - 2 = 8$

Step 5: Critical t -value (two-tailed) $|t_{critical}| = 2.306$

Compare $0.312 < 2.306$, fail to reject the null hypothesis. There is no statistically significant difference between the two sample means.

Chi-Square Test

Determines whether there is a **significant difference** between observed and expected frequencies in categorical data.

Chi-Square Tests:

Test	Description
Goodness of Fit	Compares observed vs expected frequencies for one categorical variable
Test of Independence	Determines association between two categorical variables
Test of Homogeneity	Checks if several populations have the same distribution of a categorical variable

Chi-Square Formula

Core Formula

$$\chi^2 = \sum \frac{(O - E)^2}{E}$$

- χ^2 = Chi-Square statistic
- O = Observed frequency
- E = Expected frequency
- $\sum$ = Sum over all categories/cells

Interpretation

A **larger** χ^2 value indicates a **greater difference** between observed and expected frequencies.

Z-Test

- statistical hypothesis test used to determine whether there is a significant difference between **sample** and **population** parameters
 - when the population variance is known
 - sample size is large (typically $n \geq 30$)

Z-Test	T-Test
Population variance known	Population variance un-known
Large sample ($n \geq 30$)	Small sample ($n < 30$)
Normal distribution	Normal or t-distribution

Types of Z-Tests

Type		Purpose
One-Sample Z-Test		Compare sample mean to population mean
Two-Sample Z-Test		Compare means of two independent groups
One-Proportion Test	Z-	Compare sample proportion to population proportion
Two-Proportion Test	Z-	Compare proportions of two independent groups

Examples

- Is the average height of students different from the national average?
- Do men and women have different average salaries?
- Is the proportion of voters supporting a candidate different from 50%?

Z-Test Formulas

One-Sample Z-Test (Mean)

$$Z = \frac{\bar{x} - \mu_0}{\sigma / \sqrt{n}}$$

- $\bar{x}$ = Sample mean
- μ_0 = Population mean (null hypothesis value)
- σ = Population standard deviation (known)
- n = Sample size

One-Proportion Z-Test

$$Z = \frac{\hat{p} - p_0}{\sqrt{\frac{p_0(1-p_0)}{n}}}$$

- $\hat{p}$ = Sample proportion

Two-Sample Z-Test Formulas

Two-Sample Z-Test (Means)

$$Z = \frac{(\bar{x}_1 - \bar{x}_2) - (\mu_1 - \mu_2)}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}}$$

Two-Proportion Z-Test

$$Z = \frac{(\hat{p}_1 - \hat{p}_2) - 0}{\sqrt{\hat{p}(1 - \hat{p}) \left(\frac{1}{n_1} + \frac{1}{n_2} \right)}}$$

where $\hat{p} = \frac{x_1 + x_2}{n_1 + n_2}$ (pooled proportion)

Note

Two-sample tests assume independent groups and equal variance assumptions as needed.

Example 1: One-Sample Z-Test

Problem: A manufacturer claims light bulbs last **1000 hours**. Sample of 50 bulbs has mean **980 hours**. Population $\sigma = 100$ hours. Test at $\alpha = 0.05$.

Step 1: Hypotheses

$$H_0 : \mu = 1000 \quad \text{vs} \quad H_1 : \mu \neq 1000$$

Step 2: Calculate Z

$$Z = \frac{980 - 1000}{100/\sqrt{50}} = \frac{-20}{14.142} = -1.414$$

Step 3: Critical Value

$$Z_{\text{critical}} = \pm 1.96 \quad (\alpha = 0.05, \text{two-tailed})$$

Step 4: Decision Since $|-1.414| < 1.96$, **fail to reject H_0** .

Conclusion: No evidence that mean lifetime differs from 1000 hours.

Example 2: One-Proportion Z-Test

Problem: A candidate claims **60%** support. Survey of 400 voters finds **220** supporters. Test at $\alpha = 0.05$.

Step 1: Hypotheses

$$H_0 : p = 0.60 \quad \text{vs} \quad H_1 : p \neq 0.60$$

Step 2: Sample Proportion

$$\hat{p} = \frac{220}{400} = 0.55$$

Step 3: Calculate Z

$$Z = \frac{0.55 - 0.60}{\sqrt{\frac{0.60 \times 0.40}{400}}} = \frac{-0.05}{0.0245} = -2.04$$

Step 4: Decision Since $|-2.04| > 1.96$, **reject H_0** .

Conclusion: Proportion is significantly different from 60%.

Example 3: Two-Sample Z-Test

Problem: Compare test scores between two teaching methods.

	Method A	Method B
Sample size (n)	40	35
Mean ($\bar{x}$)	78	74
Population SD (σ)	10	12

$$Z = \frac{78 - 74}{\sqrt{\frac{100}{40} + \frac{144}{35}}} = \frac{4}{\sqrt{2.5 + 4.114}} = \frac{4}{2.572} = 1.555$$

Critical value (one-tailed, $\alpha = 0.05$): 1.645

Since $1.555 < 1.645$, **fail to reject H_0** .

Conclusion: No evidence Method A is better.

Critical Values (Z-Table)

Two-Tailed Tests		
α	$Z_{\alpha/2}$	Confidence Level
0.10	1.645	90%
0.05	1.960	95%
0.01	2.576	99%
0.001	3.291	99.9%

One-Tailed Tests	
α	Z_{α}
0.10	1.282
0.05	1.645
0.01	2.326

Decision Rule

$$|Z| > Z_{\text{critical}} \rightarrow \text{Reject } H_0$$

Assumptions of Z-Test

Assumption	Requirement
Normality	Data approximately normal OR $n \geq 30$ (CLT)
Known Variance	Population standard deviation (σ) is known
Independence	Observations are independent
Random Sampling	Data collected randomly

For Proportion Tests

$$np \geq 10 \quad \text{and} \quad n(1 - p) \geq 10$$

P-Value from Z-Statistic

Z-Value	P-Value (Two-Tailed)	Significance
1.00	0.3173	Not significant
1.96	0.0500	Borderline
2.00	0.0455	Significant ($p < 0.05$)
2.58	0.0099	Highly significant
3.00	0.0027	Very highly significant

Decision Rule

$$p\text{-value} \leq \alpha \rightarrow \text{Reject } H_0$$

$$p\text{-value} > \alpha \rightarrow \text{Fail to reject } H_0$$

Effect Size: Cohen's d

Formula (One-Sample)

$$d = \frac{\bar{x} - \mu_0}{\sigma}$$

d	Effect Size
0.00 - 0.20	Negligible
0.20 - 0.50	Small
0.50 - 0.80	Medium
0.80 - 1.00	Large
> 1.00	Very large

From Bulb Example

$$d = \frac{980 - 1000}{100} = -0.20 \quad (\text{Small effect})$$

Practice Problems

- ① A company claims batteries last **50 hours**. Sample of 100 batteries has mean **48 hours** with $\sigma = 5$ hours. Test at $\alpha = 0.01$.
- ② A university claims **80%** of graduates find jobs within 6 months. Survey of 200 graduates shows **148** found jobs. Test at $\alpha = 0.05$.
- ③ Compare two manufacturing processes:
Process A: $n = 50$, $\bar{x} = 95$, $\sigma = 8$
Process B: $n = 45$, $\bar{x} = 92$, $\sigma = 10$
Test if Process A is better at $\alpha = 0.05$.

Takeaways

- 1 **Z-test** compares sample statistics to population parameters when **variance is known**
- 2 Use Z-test for **proportions** and for means with **large samples** ($n \geq 30$)
- 3 **Three main types:** One-sample, Two-sample, and Proportion tests
- 4 **Critical values** depend on α and whether test is one-tailed or two-tailed
- 5 Report both the **Z-statistic** and **p-value** for complete inference
- 6 **Effect size (Cohen's d)** tells you about practical significance
- 7 If σ is unknown and n is small, use **T-test** instead

Statistics is the tool. Data is the evidence. Decisions are the impact.